Filtered historical simulation: VaR 99% of 97 k€ for 22 Oct 2026 (1.0% of the book); 38 exceptions in 3,824 days for 38 expected

The method recommended by the backtest is selected; the slicer switches method. Each VaR is computed at the close for the next trading day.

Date

02/01/2008 — 22/10/2026

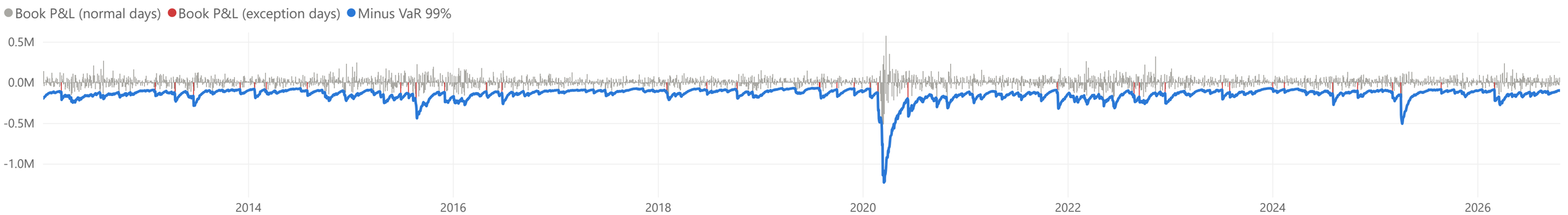
Method

Filtered historical simulation

VaR 99% (latest)	ES 97.5% (latest)	VaR 99% (latest), % of Book	Exceptions 250d	Traffic Light	Latest VaR Date
96,880	115,558	0.97%	1	Green	22 Oct 2026

Daily P&L of the book against minus the VaR of the previous close

Red days lost more than the VaR. One axis: the line is the VaR drawn as a loss.

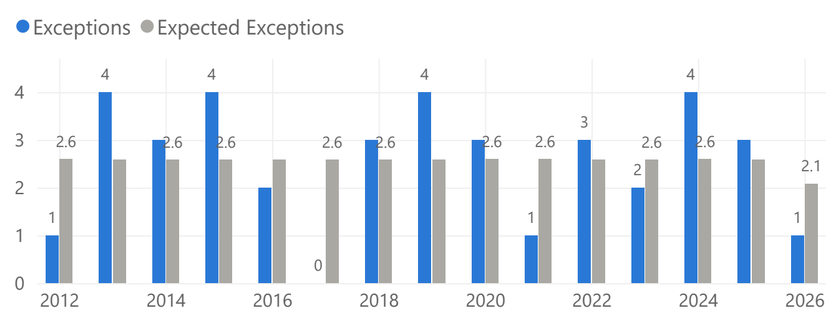


Backtest of every method, computed in Python

Tinted p-values are below 5%: the test rejects the method. Zones: Basel traffic light on 250 days.

Method	Backtest Period	Exceptions	Expected Exceptions	Kupiec p	Christoffersen p	Conditional Coverage p	Worst Zone
Filtered historical simulation	2012-2026	38	38.2	0.969	0.382	0.682	yellow
Historical simulation	2012-2026	55	38.2	0.011	0.000	0.000	red
Parametric (EWMA)	2012-2026	74	38.2	0.000	0.003	0.000	red
Parametric (GARCH)	2012-2026	56	38.2	0.007	0.010	0.001	red
Parametric (GJR)	2012-2026	54	38.2	0.016	0.214	0.025	yellow
Filtered historical simulation	2020	3	2.6	0.803	0.790	0.936	yellow

Exceptions by year against the 1% expected



GJR-GARCH(1,1) forecasts the book's volatility best (QLIKE 0.399 against 0.427 for HAR); XGBoost leads on the three equity indices

QLIKE (lower is better) and Diebold-Mariano tests against HAR, computed in Python on 2,269 out-of-sample days from 2018; GJR-GARCH feeds the VaR.

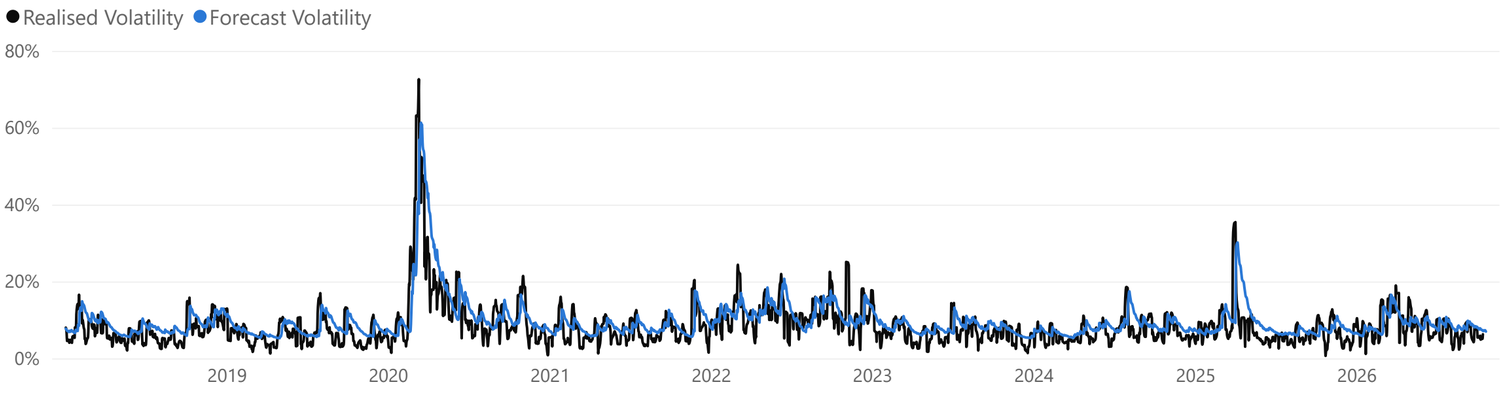
Series Book

Model GJR-GARCH(1,1)

Evaluation Period 2018-2026

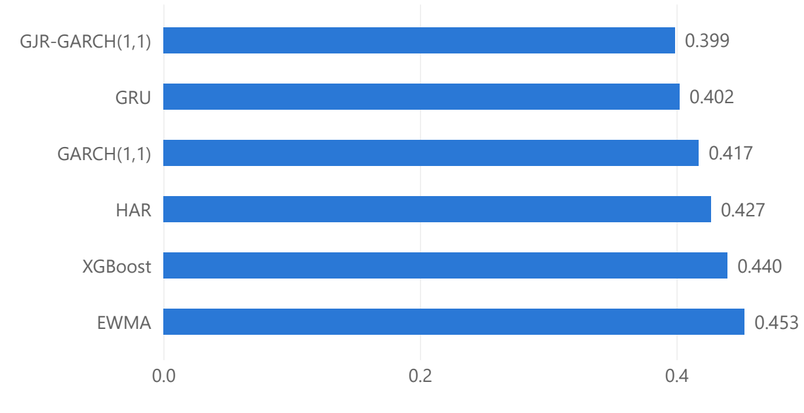
Forecast against realised volatility over the next 5 days, annualised

Series and model chosen above; forecasts made at each close from 2018.



QLIKE by model for the selected series

Lower is better.



QLIKE of every model on every series

Empty: HAR-range needs open, high, low and close, which the bond, the dollar and the book lack.

Series	EWMA	GARCH(1,1)	GJR-GARCH(1,1)	GRU	HAR	HAR (range)	XGBoost
Book	0.453	0.417	0.399	0.402	0.427		0.440
CAC 40	0.541	0.491	0.467	0.512	0.515	0.450	0.437
Euro Stoxx 50	0.534	0.494	0.464	0.503	0.518	0.466	0.452
S&P 500	0.499	0.441	0.413	0.436	0.456	0.444	0.404
Gold	0.441	0.419	0.414	0.551	0.427	0.420	0.472
10-year euro zero-coupon bond	0.339	0.332	0.332	0.391	0.344		0.391
USD cash	0.368	0.351	0.352	0.415	0.373		0.402

QLIKE by market regime, selected series

Walk-forward HMM regime, 2018-2026.

Model	calm	normal	stress
EWMA	0.456	0.406	0.470
GARCH(1,1)	0.398	0.391	0.453
GJR-GARCH(1,1)	0.404	0.378	0.401
GRU	0.415	0.331	0.419
HAR	0.407	0.410	0.458
XGBoost	0.431	0.384	0.476

Diebold-Mariano against HAR for the selected series

Negative statistic: lower losses than HAR. Tinted: significant at 5%.

Model	QLIKE ▲	DM Statistic vs HAR	DM p vs HAR
GJR-GARCH(1,1)	0.399	-1.75	0.080
GRU	0.402	-1.43	0.153
GARCH(1,1)	0.417	-1.12	0.262
HAR	0.427		
XGBoost	0.440	0.64	0.524
EWMA	0.453	1.69	0.090

XGBoost beats the 20-day ADV on all 10 stocks (25.1% mean error against 31.8%); 10% of the volume costs 35 to 60 bp on 22 Oct 2026

Volume: shares traded on Euronext Paris, as multiples of the 20-day ADV. Cost = half the estimated spread + Y × daily volatility × √(order / volume), Y = 1, estimated at the previous close.

Instrument TotalEnergies

Model XGBoost (log volume)

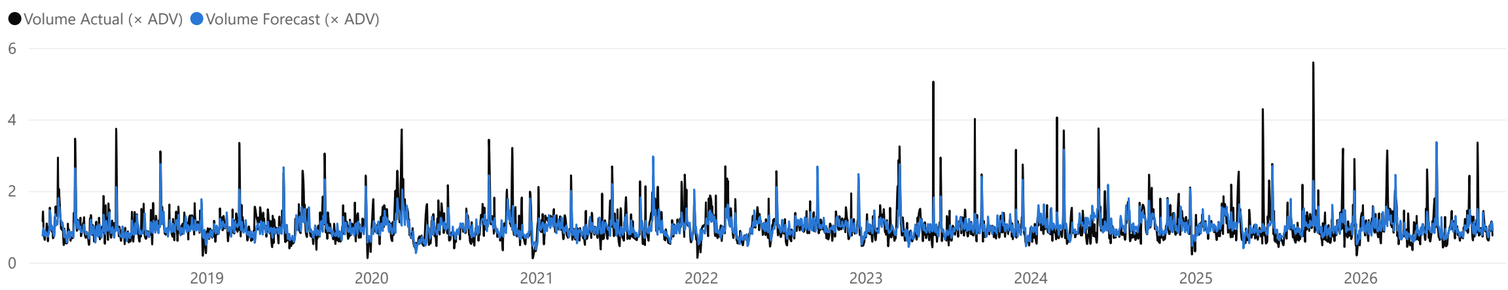
Order Size (% of volume) 5%

Date 02/01/2008 – 22/10/2026

Next Session	Next Session Volume (× ADV)	Spread, Next Session (bp)	Volatility, Next Session (bp)	Order Size	Cost at Order Size, Next Session (bp)
22 Oct 2026	0.90	11.63	119.94	5%	32.63

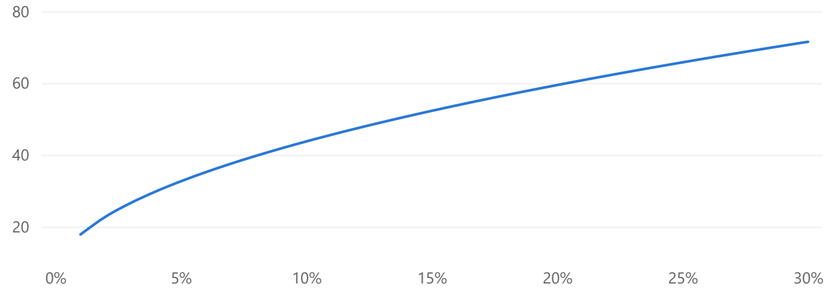
Volume of each session against the forecast, as a multiple of the 20-day ADV

Stock and model chosen above; the ADV itself forecasts 1.0. Forecasts from 2018.



Cost of an order against its size, next session

Basis points of the order's value; x: order size in % of the session's volume.



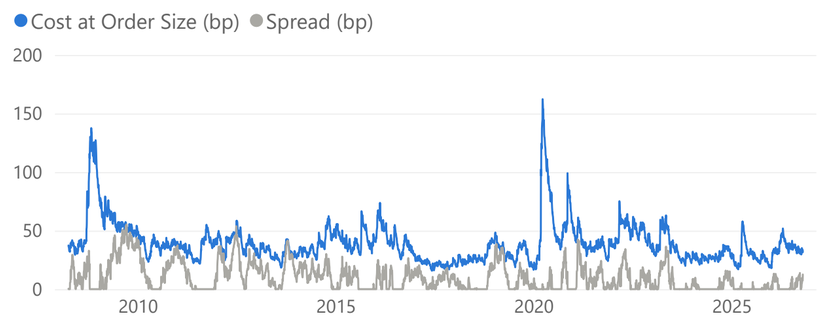
Volume models for the selected stock, computed in Python

MSE of log volume and mean absolute % error; negative statistic: better than the ADV; tinted: significant at 5%.

Model	Volume Evaluation Period	MSE (log volume)	MAPE	DM Statistic vs ADV	DM p vs ADV
20-day ADV	2018-2026	0.1358	30.4%		
HAR (log volume)	2018-2026	0.1091	25.1%	-7.30	0.0000
XGBoost (log volume)	2018-2026	0.0826	22.4%	-10.27	0.0000
20-day ADV	2020	0.1875	38.1%		
HAR (log volume)	2020	0.1195	25.3%	-3.69	0.0002
XGBoost (log volume)	2020	0.0992	22.8%	-3.80	0.0001
20-day ADV	2022	0.0984	25.3%		
HAR (log volume)	2022	0.0702	19.1%	-3.29	0.0010
XGBoost (log volume)	2022	0.0535	18.4%	-4.46	0.0000

Cost of the chosen order size and spread, in basis points

Each session uses the spread and volatility known at the previous close.



S&P 500 ×5.3 and gold ×4.7 since January 2008, euro area equities ×1.4; equity volatility at 0.6 times its 2008-2026 average

Headline figures as of 22 Oct 2026 (latest values: CAC 40 and Euro Stoxx 50 21 Oct). The snapshot, index and volatility follow the date and instrument filters.

Data as of
22 Oct 2026

Date
02/01/2008 22/10/2026

Instrument
All

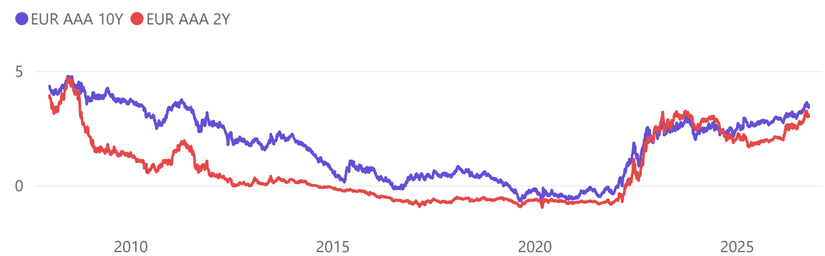
Snapshot over the selected dates

Volatility 22d vs Average below 1 means calmer than usual. Drawdowns count highs from January 2008.

Instrument	Return	Volatility (annualised)	Volatility 22d (latest)	Volatility 22d vs Average	Worst Day	Max Drawdown
S&P 500	+429.5%	19.8%	9.8%	0.60	-12.0%	-53.3%
Gold	+371.3%	18.5%	19.1%	1.13	-10.8%	-44.4%
Euro Stoxx 50	+43.2%	21.9%	12.2%	0.63	-12.4%	-58.2%
CAC 40	+42.7%	21.4%	11.8%	0.63	-12.3%	-54.6%
EUR/USD	-23.7%	8.9%	5.4%	0.66	-4.6%	-40.2%

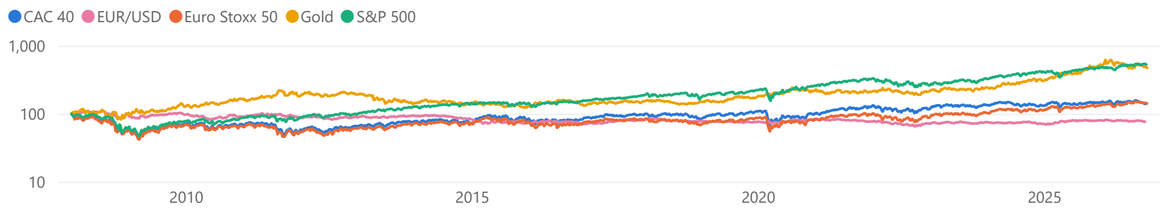
Rates, in percent

Negative at times, Jun 2016 to Mar 2022; low −0.82% (09 Mar 2020).

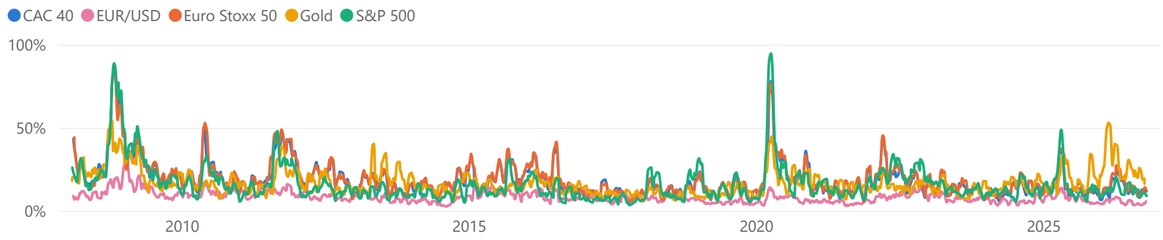


Index levels rebased to 100 on the first selected day

Logarithmic scale: equal distances are equal percentage moves.



Volatility of daily returns over the last 22 trading days, annualised



Market regime at each close: calm on 22 Oct 2026, since 22 Jun 2026

Probability of each regime from a hidden Markov model on equity volatility, return and correlation and the VIX, refitted every January on past data only.



Euro bonds no longer offset equity losses: Euro Stoxx 50 returns and 10-year yield changes correlate at −0.47 over the last 250 days, against +0.20 since 2008

Correlations use daily returns, and daily changes for the yield, on the days both series moved.

Period

Full history

Instrument A

Euro Stoxx 50

Instrument B

EUR AAA 10Y

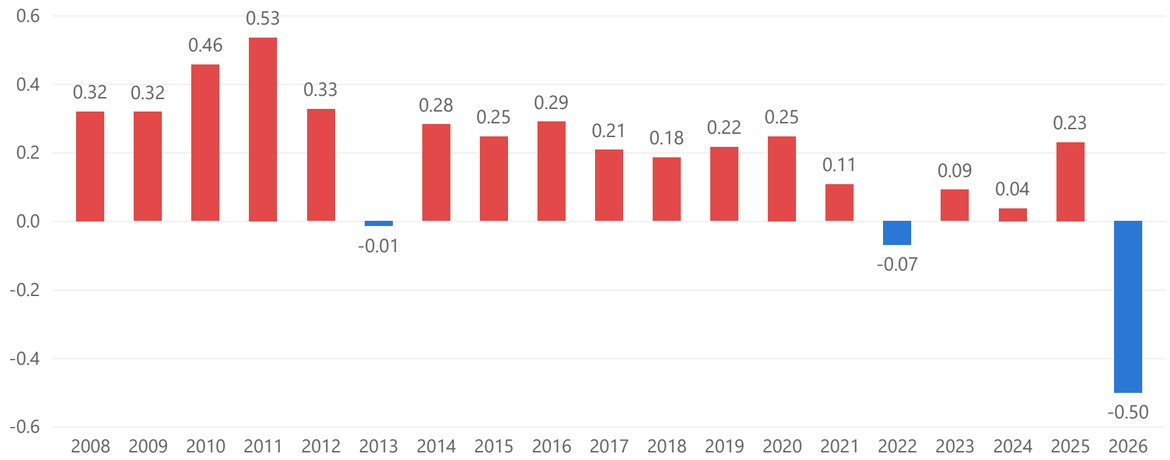
Correlation of daily moves over the selected period

Blue: the pair moves in opposite directions. Red: it moves together.

Instrument A	CAC 40	EUR AAA 10Y	EUR/USD	Euro Stoxx 50	Gold	S&P 500
CAC 40	1.00	0.21	0.17	0.97	0.02	0.58
EUR AAA 10Y	0.21	1.00	0.11	0.20	-0.12	0.09
EUR/USD	0.17	0.11	1.00	0.18	0.22	0.12
Euro Stoxx 50	0.97	0.20	0.18	1.00	0.03	0.59
Gold	0.02	-0.12	0.22	0.03	1.00	0.04
S&P 500	0.58	0.09	0.12	0.59	0.04	1.00

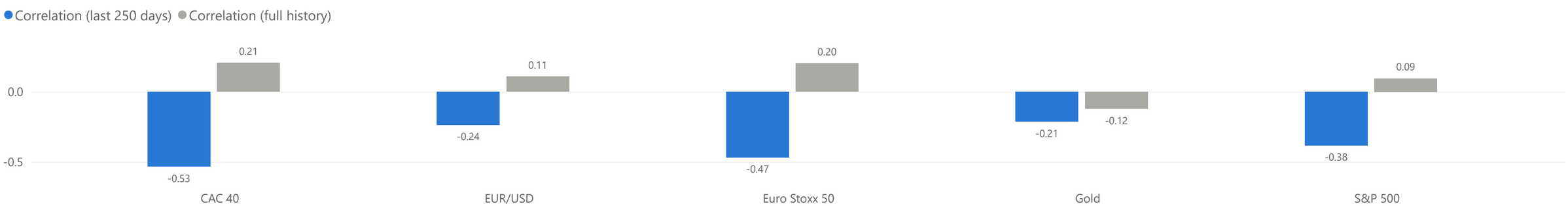
Correlation of the selected pair, by calendar year

2026 is the year to date.



Correlation with EUR AAA 10-year yield changes: last 250 days against the full history

Above zero, the bond gains when the asset falls and offsets its losses; below zero, both fall together.



Latest quality run: no critical failure; 53 warnings, 31 of them missing trading days or late series

Run of 23 Oct 2026 00:11 UTC. Rules run after each download; a critical failure stops the pipeline, a warning is reported.

Pass Rate (latest run)	Critical Failures (latest run)	Warnings (latest run)	Series Late (latest run)	Last Quality Run (UTC)
62.7%	0	53	15	23 Oct 2026 00:11

Flagged observations by rule and series, latest run

Empty: the rule does not apply to the series.

Rule	Air Liquide	Airbus	AXA	BNP Paribas	CAC 40	Danone	EUR AAA 10Y	EUR AAA 2Y	EUR/USD	EUR/USD (Yahoo)	Euro Stoxx 50	Gold	L'Oréal	LVMH	S&P 500	Sanofi	TotalEnergies	US 10Y-2Y	Vinci	VIX
duplicate_key	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
fx_cross_source										16										
large_move	0	0	0	0	0	0			0	1	0	0	0	0	0	0	0		0	
late_data	1	1	1	1	1	1	1	1	0	0	1	0	1	1	0	1	1	0	1	1
missing_trading_day	5	5	5	5	6	5	10	10	0	29	98	3	5	5	0	5	5		5	0
non_positive_price	0	0	0	0	0	0			0	0	0	0	0	0	0	0	0		0	0
schema	0																			
stale_value	0	1	0	0	0	0	0	0	0	1	0	0	3	1	0	0	1	88	1	0
suspect_tick	1	1	0	0	0	0					0	0	0	0	3	0	0		0	
yield_out_of_range							0	0												
zero_volume	2	2	2	2		2							2	2		2	4		2	

Values flagged by the latest run

Latest first; Yahoo levels are not shown, as the licence forbids publishing them.

Observation Date	Instrument	Rule	Detail
21 Oct 2026	Air Liquide	late_data	last date 2026-10-21, expected 2026-10-22 or later
21 Oct 2026	Airbus	late_data	last date 2026-10-21, expected 2026-10-22 or later
21 Oct 2026	AXA	late_data	last date 2026-10-21, expected 2026-10-22 or later
21 Oct 2026	BNP Paribas	late_data	last date 2026-10-21, expected 2026-10-22 or later
21 Oct 2026	CAC 40	late_data	last date 2026-10-21, expected 2026-10-22 or later
21 Oct 2026	Danone	late_data	last date 2026-10-21, expected 2026-10-22 or later
21 Oct 2026	EUR AAA 10Y	late_data	last date 2026-10-21, expected 2026-10-22 or later
21 Oct 2026	EUR AAA 2Y	late_data	last date 2026-10-21, expected 2026-10-22 or later
21 Oct 2026	Euro Stoxx 50	late_data	last date 2026-10-21, expected 2026-10-22 or later

How late each series is, in trading sessions

Latest first. No last date: the source delivered nothing.

Instrument	Last Date	Expected Date	Sessions Late
Air Liquide	21 Oct 2026	22 Oct 2026	1
Airbus	21 Oct 2026	22 Oct 2026	1
AXA	21 Oct 2026	22 Oct 2026	1
BNP Paribas	21 Oct 2026	22 Oct 2026	1
CAC 40	21 Oct 2026	22 Oct 2026	1
Danone	21 Oct 2026	22 Oct 2026	1
EUR AAA 10Y	21 Oct 2026	22 Oct 2026	1
EUR AAA 2Y	21 Oct 2026	22 Oct 2026	1
Euro Stoxx 50	21 Oct 2026	22 Oct 2026	1

Anomaly detector on injected errors

Share of the 200 injected errors of each type caught, and precision of the alarms.

Detector Layer	Precision	Spike	Frozen	Decimal ×10	Split	Missing	Roll Jump
Rules: stale, missing sessions	86%	0%	100%	0%	0%	100%	0%
Robust z-score	99%	80%	1%	100%	96%	1%	0%
Isolation Forest, 0.1% budget	86%	45%	1%	99%	1%	0%	0%
Isolation Forest, 1% budget	63%	87%	4%	100%	85%	3%	11%
Production: rules + z-score	91%	80%	100%	100%	96%	100%	0%

Last night run success (23 Oct 00:19 UTC, 9 min); 4 runs, 1 stopped by a failure; GJR kept as VaR champion

Jenkins runs the chain on weekday nights and the promotion on Saturdays. Return over EWMA volatility: PSI 0.06 (stable), the volatility forecast stays calibrated.

Last Night Run Status	Last Night Run Finished (UTC)	Last Night Run Duration (min)	Rows in Gold (last success)	Failed Runs	VaR Champion
SUCCESS	23 Oct 00:19	8.6	556,664	1	GJR-GARCH(1,1)

Scheduled runs, latest first

Stage reached: the failed stage when a run stopped. Critical, Warnings: failed quality rules.

Started (UTC)	Job	Status	Stage Reached	Run Duration (min)	Rows Loaded	Critical	Warnings	Detail
23 Oct 2026 00:11	nightly	SUCCESS	Publish	8.6	556,664	0	53	
23 Oct 2026 00:09	retrain	SUCCESS	VaR	1.7	556,140			kept gjr: champion gjr QLIKE 1.606 (green), challenger garch 1.637 (green)
23 Oct 2026 00:07	nightly	FAILURE	Quality gate	0.8		1	53	failed at Quality gate
22 Oct 2026 23:54	nightly	SUCCESS	Publish	10.3	546,809	0	53	

Input drift at the latest night run

PSI of the last 250 sessions against 2008-2017: stable below 0.10, significant above 0.25.

Feature	PSI (latest run)	Drift Status (latest run)
Realised variance, 22 days (log)	0.948	significant
Realised variance, 5 days (log)	0.388	significant
Realised variance, 1 day (log)	0.143	moderate
Daily return	0.120	moderate
Return over EWMA volatility	0.061	stable

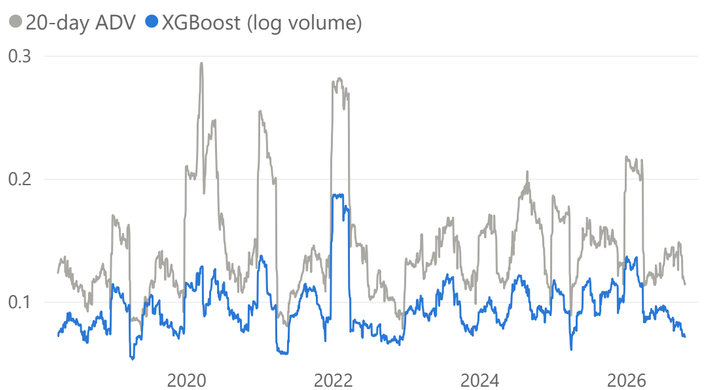
Book volatility forecast error, rolling 63 sessions (QLIKE)

Lower is better; the grey line is the baseline.



Volume forecast error, rolling 63 sessions (MSE, log)

Lower is better; the grey line is the baseline.



Champion and challenger decisions for the VaR volatility model

1-day QLIKE over 250 days; promotion needs a lower QLIKE and a zone no worse.

Decided	Champion	Champion QLIKE	Challenger	Challenger QLIKE	Promoted	After
23 Oct 2026	gjr	1.606	garch	1.637	False	gjr

Market Risk Monitor: daily risk of a 10 M€ multi-asset book

Read me: what the report shows, where the data comes from and how the figures are defined.

Purpose
Daily market risk of a fictional 10 M€ multi-asset book (CAC 40, Euro Stoxx 50, S&P 500, gold, a 10-year euro zero-coupon bond and USD cash, constant notionals), and pre-trade volume and cost estimates for ten large CAC 40 stocks, built from free data by a scheduled Python pipeline.

Pages
Risk: today's VaR and ES of the book, its exceptions and the backtest of five VaR methods.
Models: volatility forecasts against realised volatility, QLIKE and Diebold-Mariano tests, errors by regime.
Execution: next-session volume forecasts and the cost of an order, with an order-size slider.
Market: returns, volatility, drawdowns and rates of the book's assets, and the market regime.
Correlation: how the assets move together, by period.
Data quality: the quality gate, the freshness of every series and the anomaly detector.
Production: scheduled runs, input drift, rolling model errors and promotion decisions.

Data and refresh
Yahoo Finance (equity indices, gold, ten CAC 40 stocks), ECB Data Portal (EUR/USD reference rate, AAA yields) and FRED (VIX, US 10y-2y). Raw Yahoo levels and volumes are not redistributed. The report reads the gold Parquet files through the DataFolder parameter (Transform data, Edit parameters), then Refresh.
Code, notebooks and methodology: github.com/MedChahb/market_risk_monitor

Last market data

22 Oct 2026

VaR computed for

22 Oct 2026

Last quality run (UTC)

23 Oct 2026 00:11

Definitions
VaR 99%, 1 day: the loss exceeded on 1% of days, computed at each close for the next trading day.
Expected Shortfall 97.5%: the average loss on the worst 2.5% of days.
Exception: a day whose loss exceeds the VaR computed at the previous close.
Basel traffic light: exceptions over the last 250 days; green 0 to 4, yellow 5 to 9, red 10 or more.
QLIKE: loss function for variance forecasts, robust to noisy realised measures; lower is better.
Diebold-Mariano: test of equal forecast accuracy between two models.
ADV: average daily volume of the last 20 sessions. Participation: order size / session volume.
Square-root law: market impact grows with volatility and the square root of participation.
Market regime: calm, normal or stress, from a hidden Markov model on equity volatility, return, correlation and the VIX, refitted every January on past data.
Suspect tick: a move extreme for its series (robust z-score) that reverts the next day while other assets stay quiet, or larger than any real move.